EXP 6: Migrating a Web Application to Docker Containers

Theory:

Docker is a set of platform as a service (PaaS) products that use OS-level virtualization to deliver software in packages called containers. Docker is a software platform that packages software into containers. Docker images are read-only templates that contain instructions for creating a Container. A Docker image is a snapshot or blueprint of the libraries and dependencies required inside a container for an application to run. A Docker container image is a lightweight, standalone, executable package of software that includes everything needed to run an application: code, runtime

Steps for execution:

Step 1: Create EC2 Instance

- Go to AWS Management Console.
- Open **EC2 Dashboard**.
- Click **Launch Instance**.
- Choose:
  - **Linux AMI**
  - Enable **HTTP (80)** and **HTTPS (443)**
- Keep remaining settings as default.
- Click **Launch Instance**.

Step 2: Connect to EC2 Instance

- Select the created instance.
- Click **Connect**.
- Open the **Linux terminal**.

Step 3: Install Docker

- Update packages:
  sudo yum update -y
- Install Docker:
  sudo yum install docker -y
- Start Docker service:
  sudo service docker start
- Check Docker status:
  sudo service docker status

Step 4: Verify Docker Installation

- Switch to root user:
  sudo su
- Check Docker version:
  docker version

Step 5: Pull Docker Image

- Pull Nginx image:
  docker pull nginx
- List Docker images:
  docker images

Step 6: Run Docker Container

- Run Nginx container:
  docker run -d -p 80:80 nginx
- Check running containers:
  docker ps

Step 7: Access Application

- Go to EC2 instance dashboard.
- Copy the **Public IP address**.
- Paste it in a web browser.
- Verify that the **Nginx web page** is displayed.

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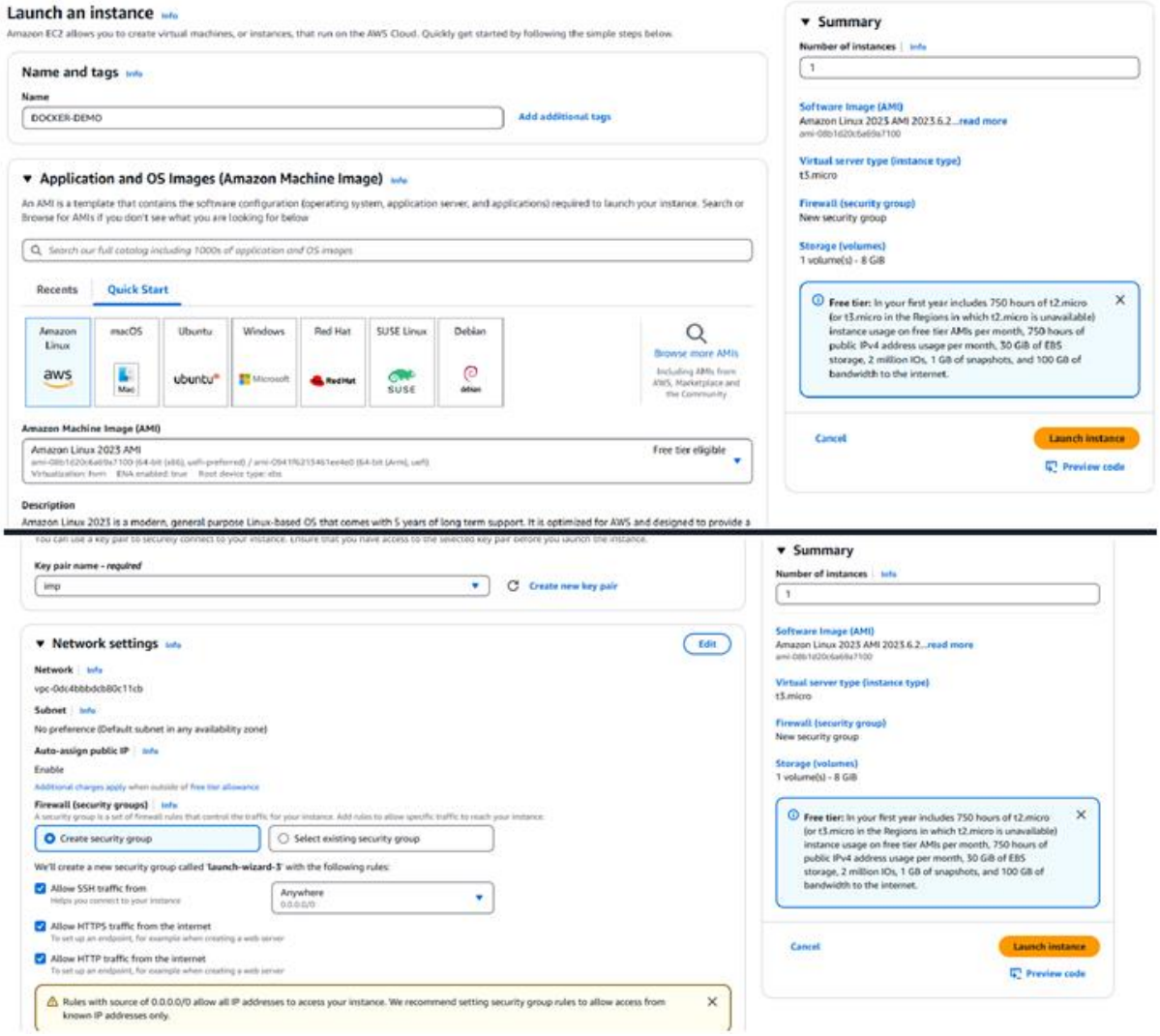

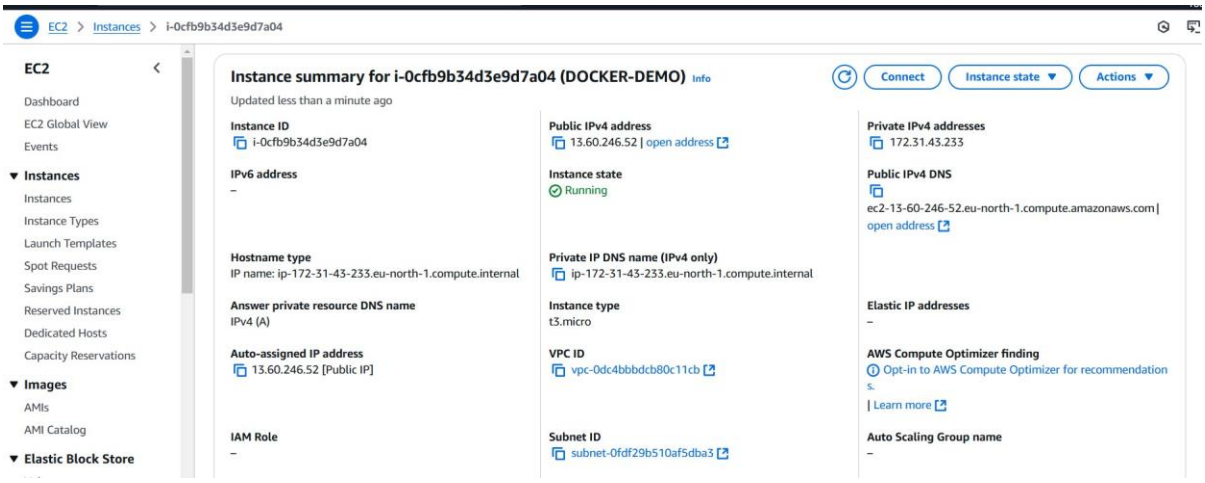

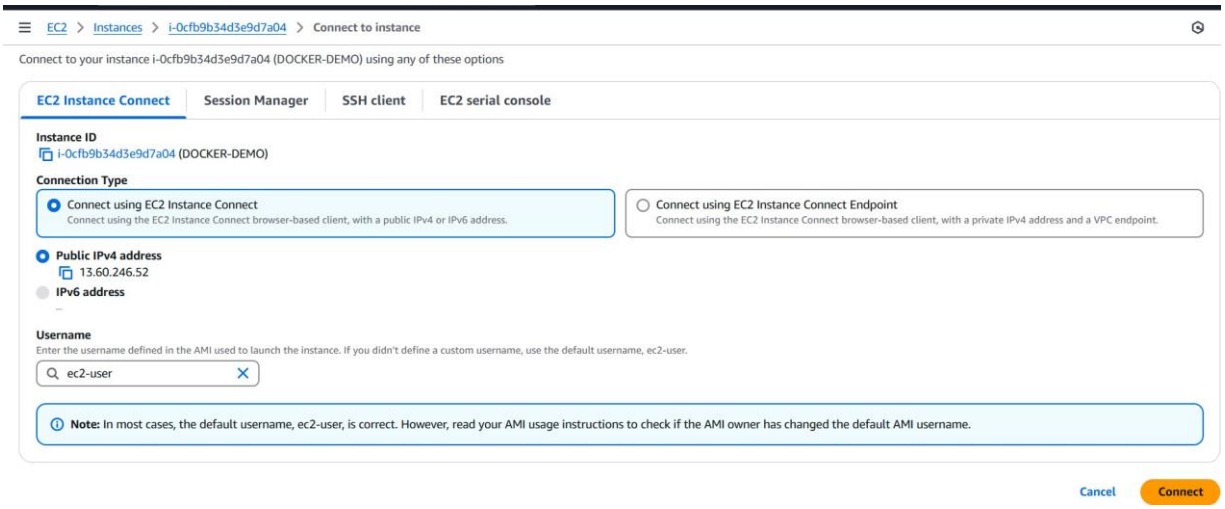
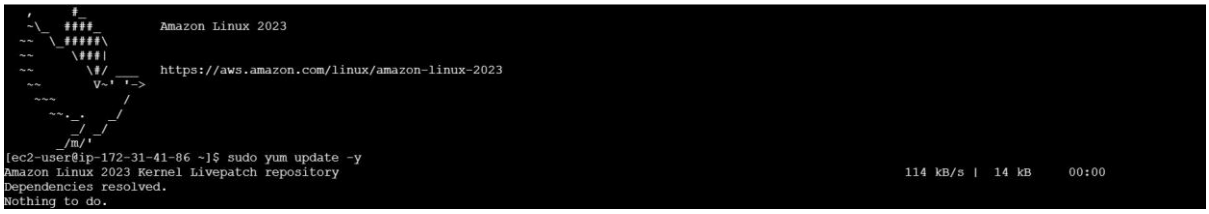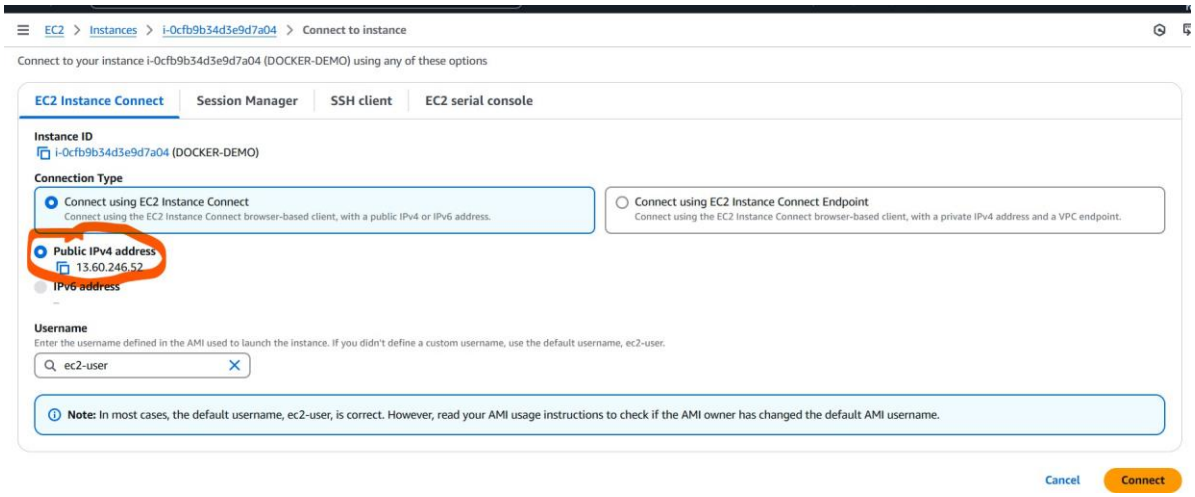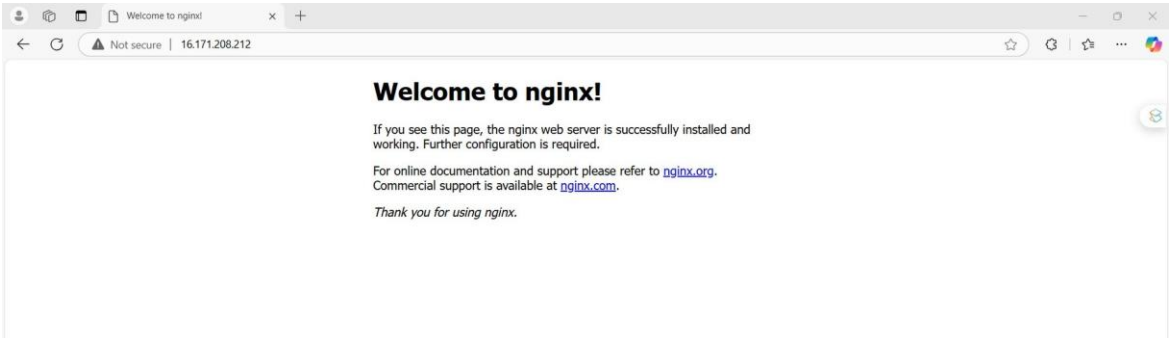
IT WILL TAKE YOU LINUX MACHINE